

Course Code: CS-11303	Course Title: Operating System
Credit Hours: 4 (3,1)	Lecture Hours/Week: 3
Batch: Fall-2025	Pre-Requisites: None

WEEK	Lab#	Lab Content
01	01	Boot options and Different Flavors of Linux Legacy Boot UEFI Open Source Close Source Ubuntu Red Hat Open Suse Fedora Kali Mint
02	02	Linux Installation and Hands on Practices History of Unix Free Software vs Open Source Open-Source Licenses History of Linux Linux Kernel Why Linux? Linux Market Share Virtual Machine Red Hat Image Layer Operating System Root User
03	03	Directory Structure Manage Directories and Files /bin /root /var /proc /sys /sbin /lib /boot /media /dev /opt /etc /srv /home /tmp
04	04	Managing Files and Directories Change Directory Contents List Directory Contents PWD Create and View Files Touch View a file with cat View a file with Less Head and Tail Copy files and make a new Directory

		Delete Files and Directories Hard vs Soft
05	05	Finding Stuff in Linux Graphical Search Tools Find Locate Which Type Grep
06	06	Linux Shells Types of Shells Login Shells Non login Shells Shell History Switching to User root Aliases
07	07	Manage Users and Groups Gid Webedit Owner ships Permissions Regular Users System Users Shadow Useradd Manage accounts and Passwords Delete User Modify User Add Groups Delete Groups Manage groups and Passwords Newgrp
08	08	Manage File Permissions and Ownerships Understanding File Permissions Types of Permissions Permission Groups Chmod Chown Chgrp Access Permissions
09	09	Mid Term Exam
10	10	Administer Linux with YAST Help resources MAN Getting to know yast Ncurses QT Man pages Info pages GUI based Help Help on the Web
11	11	Processes and Services Network Configurations and Monitoring / implementation of Memory Allocation Algorithms First Fit Next Fit

		Worst Fit Program Jobs and Processes Services Manage Foreground and Background process Process Switching Prioritize Processes Setup Network Interfaces Display the current network configuration IP address setup Device attributes Background Detailed understanding about the concept and flow Pseudo code Implementation
12	12	Implementation of CPU Scheduling Algorithms Background Detailed understanding about the concept and flow Pseudo code Implementation
13	13	Implementation of Memory Page Replacement Algorithms First-Part Background Detailed understanding about the concept and flow Pseudo code Implementation
14	14	Implementation of Memory Page Replacement Algorithms Second Part Background Detailed understanding about the concept and flow Pseudo code Implementation
15	15	Implementation of Dining Philosopher's Problem & Readers/Writers Problem Background Detailed understanding about the concept and flow Pseudo code Implementation
16	16	*Contents will be revised. Administer Linux with YAST Help resources MAN Processes and Services Network Configurations and Monitoring
		Final Term Exam